

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 1 – Pattern Recognition and Text Processing

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	Assessment:	Assessment Tool 1 – Pattern Recognition and Text Processing
Weightage:	40% of CIA		
CO5 Statement:	Analyse regular expression patterns. (BL-3)		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Code of Conduct

P. Muni Karthik / 192425061 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: P. Muni Karthik

Instructions

- Show all intermediate steps, calculations, state transitions, and reasoning wherever applicable.
- Use only the Python re (Regular Expressions) module to solve the given problems.
- Clearly mention the Regular Expression (Regex) pattern used before writing the corresponding Python program.
- Display the required output for each question, including extracted values, validation results, counts, and positions wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Regular Expression Pattern Generation	Pattern Matching Information Extraction	1
Search for a String	Keyword Search and Text Analysis	2
Split the documentation into sentences and words	Text Tokenization and Sentence Segmentation	3
Email and Strong Password Validation	Data Validation and Format Verification	4
Compile Once, Validate Many	Pattern Reusability and Performance Optimization	5

Section 1: Regular Expression Pattern Generation

Student Details:

Name: John Doe

Email: john.doe123@gmail.com

Mobile: +91-9876543210

Password: P@ssw0rd123

Date of Birth: 15/08/2004

Register Number: 23AIML1056

Department: Artificial Intelligence and Machine Learning

For the given student details formulate Regex patterns for the following:

Email ID

Mobile Number

Strong Password

Date of Birth (DD/MM/YYYY)

Register Number

Section 2: Search for a String

Artificial Intelligence (AI) is transforming industries across the world. AI is used in healthcare to assist doctors in diagnosis, in banking to detect fraud, and in education to provide personalized learning experiences. Many companies invest heavily in AI research because AI improves efficiency and enables intelligent decision-making. As AI continues to evolve, professionals with AI skills are in high demand.

Write a Python program using the re module to perform the following operations for the word "AI":

1. Find the first occurrence of the word "AI" using re.search().
2. Display the starting position of the first occurrence.
3. Display the ending position of the first occurrence.
4. Display the total number of times the word "AI" appears in the passage.
5. Print an appropriate message if the word is not found.

Section 3: Split the documentation into sentences and words

Refer the previous passage and Write a Python program using the re.split() method to perform the following operations:

- Split the passage into sentences using appropriate punctuation (., ?, !) as delimiters.
- Count and display the total number of sentences.
- Split the passage into words by considering one or more whitespace characters as delimiters.
- Count and display the total number of words.
- Display the list of extracted sentences and words.

Section 4: Email and Strong Password Validation

A website requires users to register by providing an Email ID and a Strong Password. Write a Python program using the re module to validate the following inputs.

Validation Criteria:

Email ID

- Must begin with one or more letters or digits.
- May contain ., _, %, +, or - before the @ symbol.
- Must contain a valid domain name.
- Must end with a valid domain extension such as .com, .org, .edu, or .in.

Strong Password

- Must contain at least 8 characters.
- Must contain at least one uppercase letter.
- Must contain at least one lowercase letter.
- Must contain at least one digit.
- Must contain at least one special character from @, #, \$, %, &, !, or *.
- Must not contain any whitespace characters.

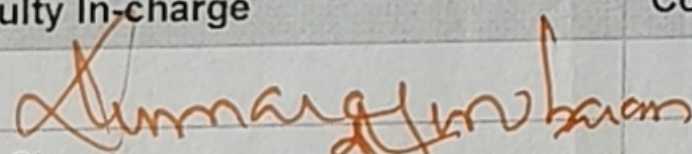
Section 5: Compile Once, Validate Many

A company receives 1,000 email IDs from its customers and needs to validate each one efficiently. Instead of compiling the same regular expression for every email, write a Python program using the re.compile() method to compile the pattern once and reuse it to validate all the email IDs.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Q. No. / Criteria	Conceptual Understanding	Accuracy of Answer	Application of Knowledge	Completeness of Response	Clarity & Presentation	Sub. Total	Total %
1	3	3	3	4	4	17	83
2						17	
3						17	
4						14	
5						14	

Faculty In-charge	Course Coordinator	Program Director
		
Signature: _____	Signature: _____	Signature: _____
Date: <u>28/7</u>	Date: _____	Date: _____

NAME OF THE STUDENT : P. MUNI KARTHIK

REGISTER NUMBER : 192425061

DEPARTMENT : AIML

SLOT : C

COURSE CODE : DSA0306

COURSE NAME : NATURAL LANGUAGE PROCESSING

COURSE FACULTY : Dr. KUMARAGURUBARAN. T

ASSESSMENT TOOL NO : 01

CO1
ASSESSMENT TOOL 1
Pattern Recognition and Text Processing

Section 1: Regular Expression Pattern Generation

Student Details:

Name: John Doe

Email: john.doe123@gmail.com

Mobile: +91-9876543210

Password: P@ssw0rd123

Date of Birth: 15/08/2004

Register Number: 23AIML1056

Department: Artificial Intelligence and Machine Learning

For the given student details formulate Regex patterns for the following:

Email ID

Mobile Number

Strong Password

Date of Birth (DD/MM/YYYY)

Register Number

PYTHON CODE:

```
import re
name = input("Enter Name: ")
email = input("Enter Email ID: ")
mobile = input("Enter Mobile Number: ")
password = input("Enter Password: ")
dob = input("Enter Date of Birth (DD/MM/YYYY): ")
regno = input("Enter Register Number: ")

email_pattern = r'^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$'
mobile_pattern = r'^[6-9]\d{9}$'
password_pattern = r'^(?=.*[a-z])(?=.*[A-Z])(?=.*\d)(?=.*[@#%&+=!]).{8,}$'
dob_pattern = r'^(0[1-9]|12)[0-9]/(0[1-9]|1[0-2])/d{4}$'
reg_pattern = r'^\d{2}[A-Z]{4}\d{4}$'

print("\n----- Validation Result -----")
if re.fullmatch(email_pattern, email):
    print("Email ID : Valid")
else:
    print("Email ID : Invalid")
```



```

if re.fullmatch(mobile_pattern, mobile):
    print("Mobile Number : Valid")
else:
    print("Mobile Number : Invalid")

```

```

if re.fullmatch(password_pattern, password):
    print("Strong Password : Valid")
else:
    print("Strong Password : Invalid")

```

```

if re.fullmatch(dob_pattern, dob):
    print("Date of Birth : Valid")
else:
    print("Date of Birth : Invalid")

```

```

if re.fullmatch(reg_pattern, regno):
    print("Register Number : Valid")
else:
    print("Register Number : Invalid")

```

output:

The image shows a screenshot of a Windows desktop with two windows. The left window is a text editor containing a Python script for user registration validation. The right window is a terminal showing the execution of the script with sample input and output.

Python Script (Left Window):

```

name = input("Enter Name : ")
email = input("Enter Email : ")
mobile = input("Enter Mobile Number : ")
password = input("Enter Password : ")
dob = input("Enter Date of Birth (DD/MM/YYYY) : ")
regno = input("Enter Registered Number : ")

# Regular Expression Patterns
email_pattern = r'[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$'
mobile_pattern = r'[6-9][0-9]{9}$'
password_pattern = r'^(?=.*[a-z])(?=.*[A-Z])(?=.*\d)(?=.*[@$!%*^&])[a-zA-Z\d@$!%*^&]{8,}$'
dob_pattern = r'^([0-1]?[0-9]|1[0-12])/([0-1]?[0-9]|1[0-2])/([0-1]?[0-9]|1[0-4])$'
reg_pattern = r'^[0-9]{10}$'

print("\nValidation Results")

if re.fullmatch(email_pattern, email):
    print("Email ID : Valid")
else:
    print("Email ID : Invalid")

if re.fullmatch(mobile_pattern, mobile):
    print("Mobile Number : Valid")
else:
    print("Mobile Number : Invalid")

if re.fullmatch(password_pattern, password):
    print("Strong Password : Valid")
else:
    print("Strong Password : Invalid")

if re.fullmatch(dob_pattern, dob):
    print("Date of Birth : Valid")
else:
    print("Date of Birth : Invalid")

if re.fullmatch(reg_pattern, regno):
    print("Registered Number : Valid")
else:
    print("Registered Number : Invalid")

```

Terminal Output (Right Window):

```

Python 3.11.5 (tags/v3.11.5:6b2b142, Jun 11 2023, 19:15:46) [AMD64] on
win32
Enter "help" below or click "help" above for more information.

>>>
Enter Name: madhu
Enter Email ID: madhu.mee123@gmail.com
Enter Mobile Number: 9744445552
Enter Password: Madhu@123
Enter Date of Birth (DD/MM/YYYY): 25/05/2004
Enter Registered Number: 190425002

----- Validation Result -----
Email ID : Valid
Mobile Number : Valid
Strong Password : Valid
Date of Birth : Valid
Registered Number : Valid

>>>

```

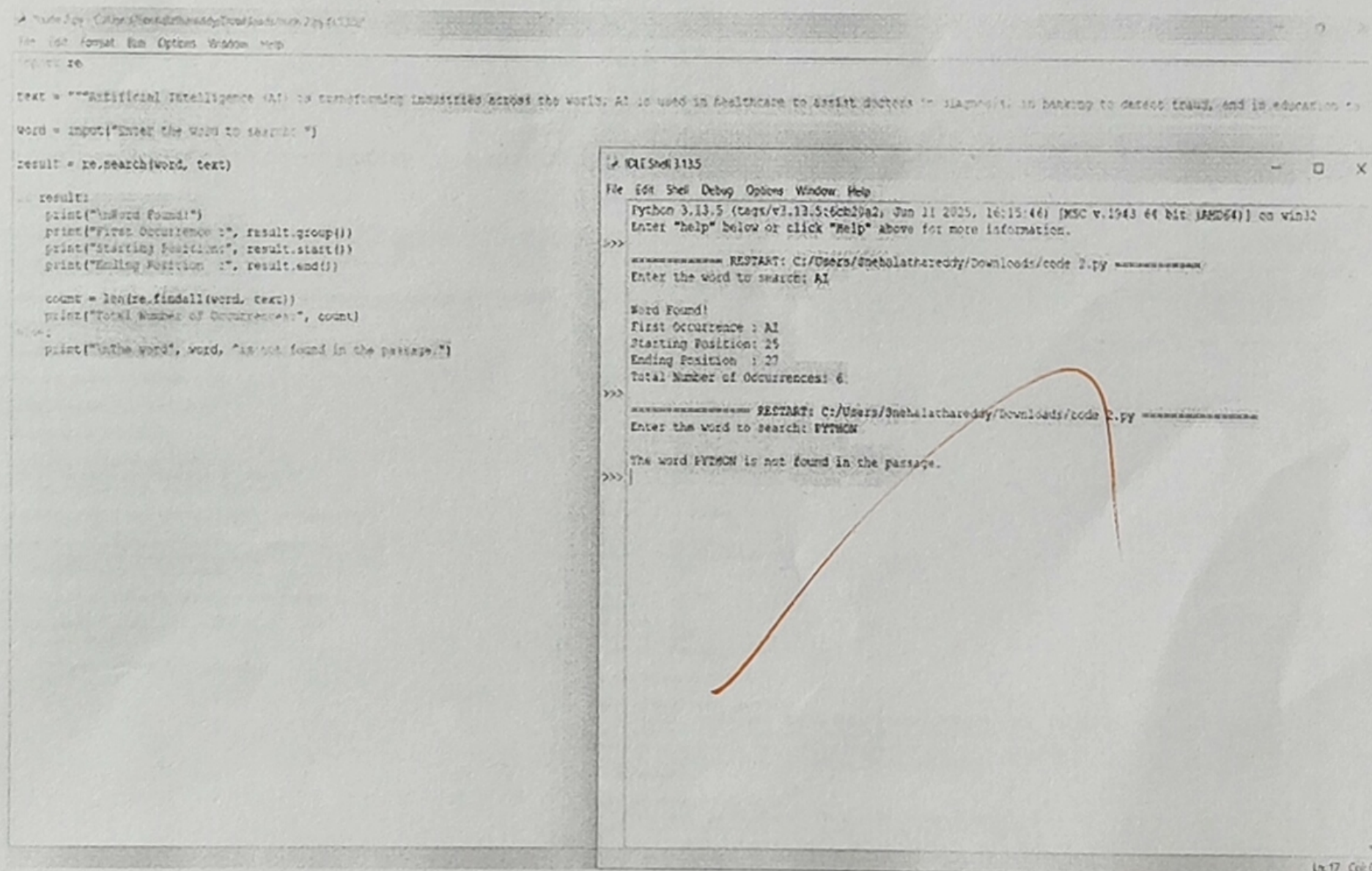

Section 2: Search for a String

Artificial Intelligence (AI) is transforming industries across the world. AI is used in healthcare to assist doctors in diagnosis, in banking to detect fraud, and in education to provide personalized learning experiences. Many companies invest heavily in AI research because AI improves efficiency and enables intelligent decision-making. As AI continues to evolve, professionals with AI skills are in high demand.

Write a Python program using the re module to perform the following operations for the word "AI":

1. Find the first occurrence of the word "AI" using re.search().
2. Display the starting position of the first occurrence.
3. Display the ending position of the first occurrence.
4. Display the total number of times the word "AI" appears in the passage.

Print an appropriate message if the word is not found.



```
code 2.py - C:\Users\Shelathareddy\Downloads\code 2.py (Python 3.11.5)
File Edit Format Run Options Window Help

import re

text = """Artificial Intelligence (AI) is transforming industries across the world. AI is used in healthcare to assist doctors in diagnosis, in banking to detect fraud, and in education to provide personalized learning experiences. Many companies invest heavily in AI research because AI improves efficiency and enables intelligent decision-making. As AI continues to evolve, professionals with AI skills are in high demand."""

word = input("Enter the word to search: ")

result = re.search(word, text)

if result:
    print("Word Found!")
    print("First Occurrence:", result.group())
    print("Starting Position:", result.start())
    print("Ending Position:", result.end())

    count = len(re.findall(word, text))
    print("Total Number of Occurrences:", count)

    print("The word", word, "is found in the passage.")
else:
    print("The word", word, "is not found in the passage.")

Python 3.11.5 (tags/v1.13.5:6b2b9a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "help" above for more information.

===== RESTART: C:\Users\Shelathareddy\Downloads\code 2.py =====
Enter the word to search: AI

Word Found!
First Occurrence: AI
Starting Position: 26
Ending Position: 27
Total Number of Occurrences: 6

===== RESTART: C:\Users\Shelathareddy\Downloads\code 2.py =====
Enter the word to search: PYTHON

The word PYTHON is not found in the passage.

Ln 17 Col 0
```


Section 3: Split the documentation into sentences and words

Refer the previous passage and Write a Python program using the `re.split()` method to perform the following operations:

- Split the passage into sentences using appropriate punctuation (., ?, !) as delimiters.
- Count and display the total number of sentences.
- Split the passage into words by considering one or more whitespace characters as delimiters.
- Count and display the total number of words. Display the list of extracted sentences and words.

OUTPUT:

```
Python 3.11.3 Shell
File Edit Shell Debug Options Window Help
Python 3.11.3 (tags/v3.11.3:686bcb5, Jun 13 2023, 16:13:14) [AMD64] on win32
Enter "help" below or click "help" above for more information.

>>>
----- EXTRACTED SENTENCES -----
Sentence 1 : Artificial Intelligence (AI) is transforming industries across the world.
Sentence 2 : AI is used in healthcare to assist doctors in diagnosis, in banking to detect fraud, and in education to provide personalized learning experiences.
Sentence 3 : Many companies invest heavily in AI research because AI improves efficiency and enables intelligent decision-making.
Sentence 4 : As AI continues to evolve, professionals with AI skills are in high demand.

Total Number of Sentences: 4

----- EXTRACTED WORDS -----
Word 1 : Artificial
Word 2 : Intelligence
Word 3 : AI
Word 4 : is
Word 5 : transforming
Word 6 : industries
Word 7 : across
Word 8 : the
Word 9 : world.
Word 10 : AI
Word 11 : is
Word 12 : used
Word 13 : in
Word 14 : healthcare
Word 15 : to
Word 16 : assist
Word 17 : doctors
Word 18 : in
Word 19 : diagnosis.
Word 20 : in
Word 21 : banking
Word 22 : to
Word 23 : detect
Word 24 : fraud.
Word 25 : and
Word 26 : in
Word 27 : education
Word 28 : to
Word 29 : provide
Word 30 : personalized
Word 31 : learning
Word 32 : experiences.
Word 33 : Many
Word 34 : companies
Word 35 : invest
Word 36 : heavily
Word 37 : in
Word 38 : AI
Word 39 : research
Word 40 : because
Word 41 : AI
Word 42 : improves
Word 43 : efficiency
Word 44 : and
Word 45 : enables
Word 46 : intelligent
Word 47 : decision-making.
Word 48 : As
Word 49 : AI
Word 50 : continues
Word 51 : to
Word 52 : evolve.
Word 53 : professionals
Word 54 : with
Word 55 : AI
Word 56 : skills
Word 57 : are
Word 58 : in
Word 59 : high
Word 60 : demand.

Total Number of Words: 60

List of Extracted Sentences:
['Artificial Intelligence (AI) is transforming industries across the world.', 'AI is used in healthcare to assist doctors in diagnosis, in banking to detect fraud, and in education to provide personalized learning experiences.', 'Many companies invest heavily in AI research because AI improves efficiency and enables intelligent decision-making.', 'As AI continues to evolve, professionals with AI skills are in high demand.']

List of Extracted Words:
['Artificial', 'Intelligence', '(AI)', 'is', 'transforming', 'industries', 'across', 'the', 'world.', 'AI', 'is', 'used', 'in', 'healthcare', 'to', 'assist', 'doctors', 'in', 'diagnosis.', 'in', 'banking', 'to', 'detect', 'fraud.', 'and', 'in', 'education', 'to', 'provide', 'personalized', 'learning', 'experiences.', 'Many', 'companies', 'invest', 'heavily', 'in', 'AI', 'research', 'because', 'AI', 'improves', 'efficiency', 'and', 'enables', 'intelligent', 'decision-making.', 'As', 'AI', 'continues', 'to', 'evolve.', 'professionals', 'with', 'AI', 'skills', 'are', 'in', 'high', 'demand.']
```


Section 4: Email and Strong Password Validation

A website requires users to register by providing an Email ID and a Strong Password. Write a Python program using the re module to validate the following inputs.

Validation Criteria:

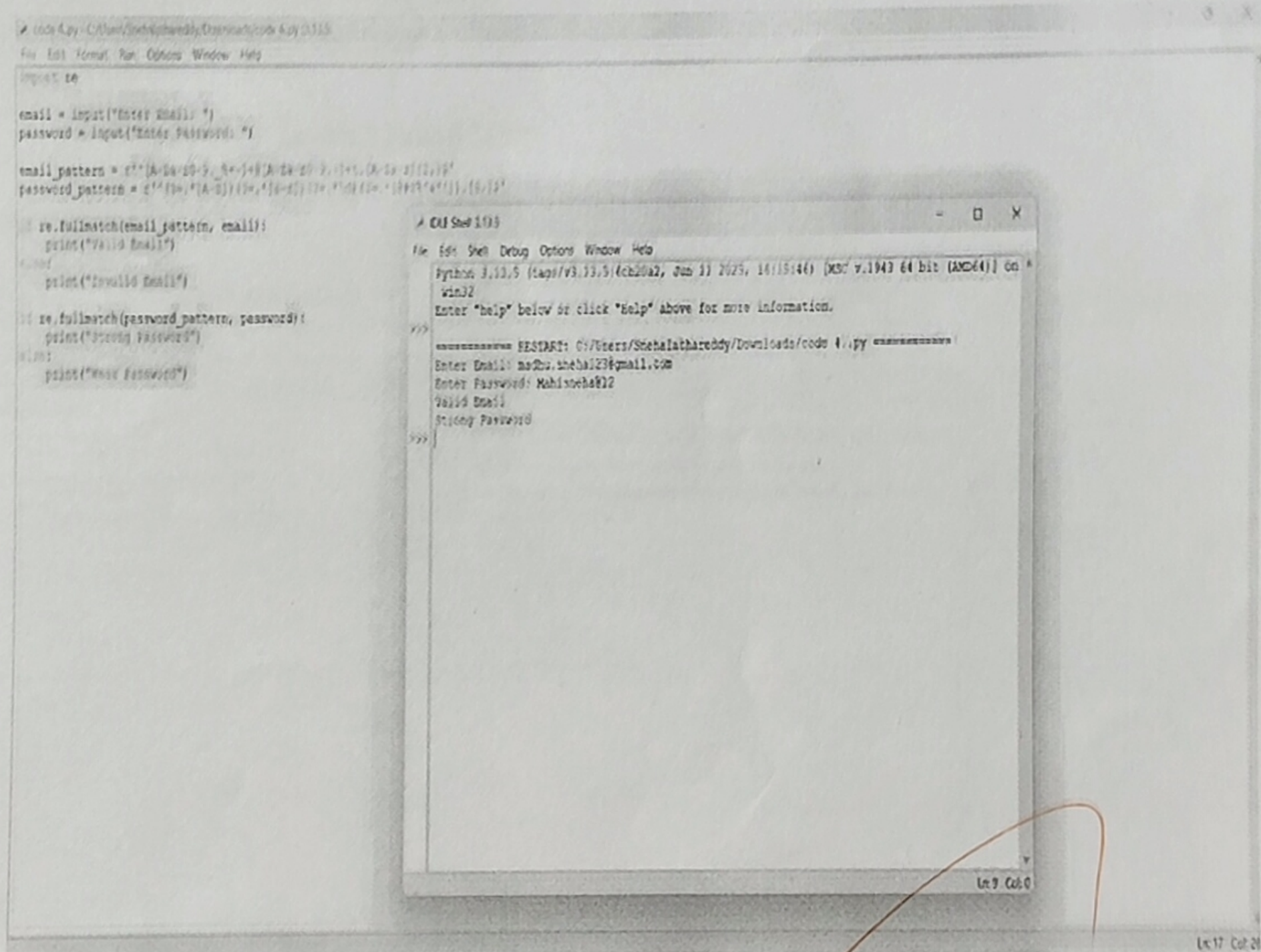
Email ID

- Must begin with one or more letters or digits.
- May contain ., _, %, +, or - before the @ symbol.
- Must contain a valid domain name.
- Must end with a valid domain extension such as .com, .org, .edu, or .in.

Strong Password

- Must contain at least 8 characters.
- Must contain at least one uppercase letter.
- Must contain at least one lowercase letter.
- Must contain at least one digit.
- Must contain at least one special character from @, #, \$, %, &, !, or *. Must not contain any whitespace characters.

OUTPUT:



The screenshot shows a Python IDE with two windows. The left window displays the Python code for validating an email and a strong password. The right window shows the output of the program.

```
code4.py - C:\Users\Shaharathreddy\Downloads\code4.py (333)
File Edit Format Run Options Window Help

Input: 26

email = input("Enter Email: ")
password = input("Enter Password: ")

email_pattern = r'^[A-Za-z0-9_+-%@]{1,100}\.[A-Za-z0-9]{1,100}\.[A-Za-z]{2,10}$'
password_pattern = r'^(?=.*[A-Z])(?=.*[a-z])(?=.*[0-9])(?=.*[!@#$%^&*])(?![\s]).{8,}$'

re.fullmatch(email_pattern, email):
print("Valid Email")
else:
print("Invalid Email")

re.fullmatch(password_pattern, password):
print("Strong Password")
else:
print("Weak Password")
```

Output:

```
C:\Shell 1.0.0
File Edit Shell Debug Options Window Help
Python 3.12.5 (tags/v3.12.5:4cb20a2, Jun 11 2025, 16:15:46) [MSVC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

===== RESTART: C:\Users\Shaharathreddy\Downloads\code4.py =====
Enter Email: ms2hu.shaharathreddy@gmail.com
Enter Password: Mahinshah@12
Valid Email
Strong Password
```


Section 5: Compile Once, Validate Many

A company receives 1,000 email IDs from its customers and needs to validate each one efficiently. Instead of compiling the same regular expression for every email, write a Python program using the `re.compile()` method to compile the pattern once and reuse it to validate all the email IDs.

PYTHON CODE:

```
import re

email_pattern = re.compile(r'^[A-Za-z0-9][A-Za-z0-9._%+-]*@[A-Za-z0-9-]+\.(com|org|edu|in)$')

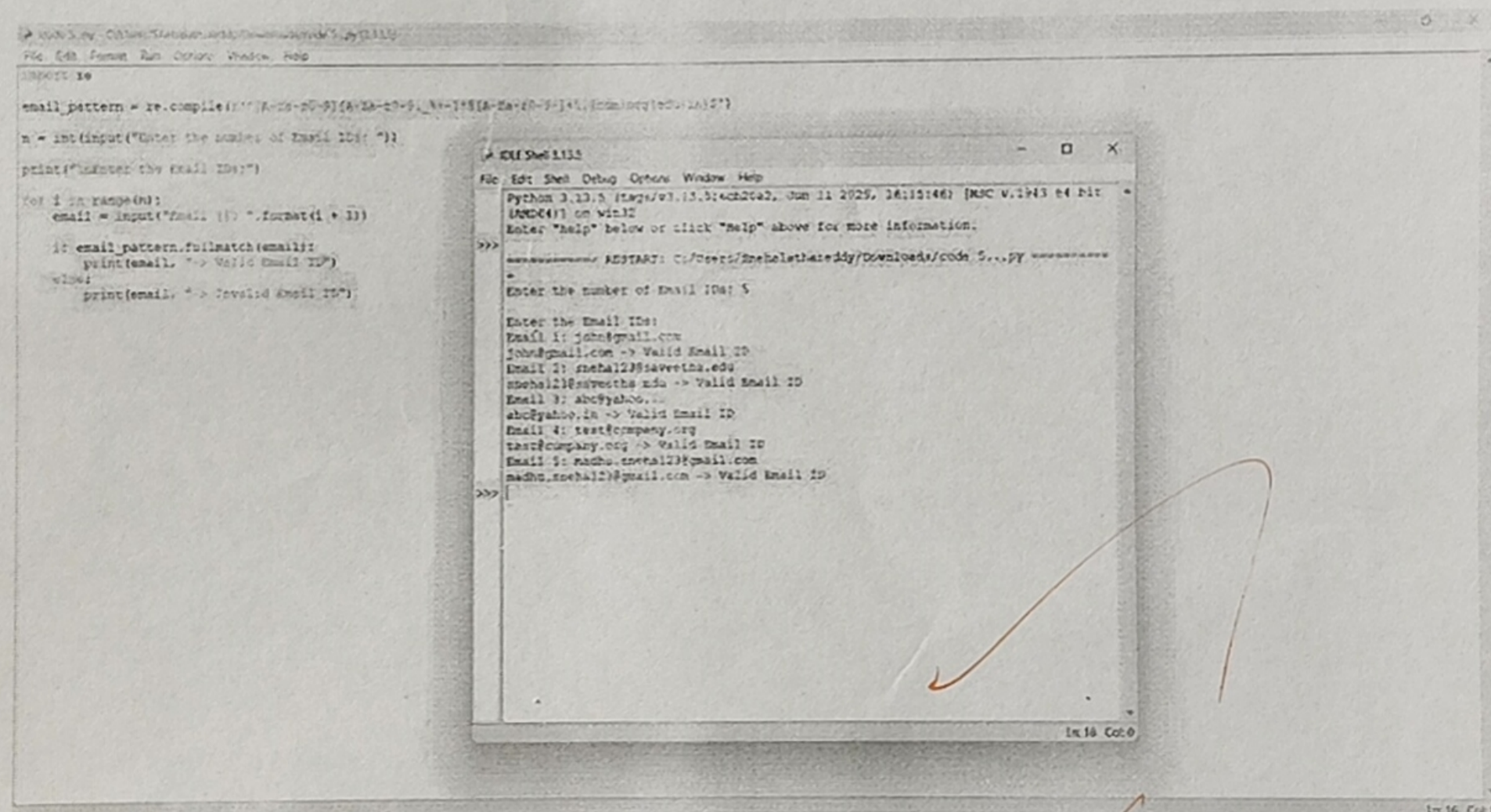
n = int(input("Enter the number of Email IDs: "))

print("\nEnter the Email IDs:")

for i in range(n):
    email = input("Email {}: ".format(i + 1))

    if email_pattern.fullmatch(email):
        print(email, "-> Valid Email ID")
    else:
        print(email, "-> Invalid Email ID")
```

OUTPUT:



```
Python 3.10.5 Shell
File Edit Shell Debug Options Window Help
Python 3.10.5 (tags/v3.10.5:2c3624e, Jun 11 2023, 16:10:46) [MSC v.1915 64-bit]
Enter "help" below or click "help" above for more information.
>>>
===== RESTART: C:\Users\Shubham\Downloads\code 5...py =====
>>>
Enter the number of Email IDs: 5
Enter the Email IDs:
Email 1: john@gmail.com
john@gmail.com -> Valid Email ID
Email 2: snehal23@savetna.edu
snehal23@savetna.edu -> Valid Email ID
Email 3: abc@yahoo...
abc@yahoo.in -> Valid Email ID
Email 4: test@company.org
test@company.org -> Valid Email ID
Email 5: madhu.snehal23@gmail.com
madhu.snehal23@gmail.com -> Valid Email ID
>>>
```